

Reinforcement Learning for Lander Control

AE4350: Bio-inspired Intelligence and Learning for Aerospace Applications

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Abbreviations

Abbreviation	Definition
ACD	Adaptive Critic Design
ANN	Artificial Neural Network
DOF	Degree of Freedom
DP	Dynamic Programming
GAE	Generalized Advantage Estimation
KL	Kullback-Leibler
MC	Monte Carlo
MDP	Markov Decision Process
MLP	Multi-Layer Perceptron
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
RMS	Root Mean Square
SAC	Soft Actor-Critic
TD	Temporal Difference
TRPO	Trust Region Policy Optimization

Symbols

Symbol	Definition	Unit
$a_t, \mathbf{u}_t$	Control action vector at step t	[-]
$\hat{A}_t$	Advantage function estimate	[-]
c_{ent}	Policy entropy bonus coefficient	[-]
c_v	Value function loss weighting coefficient	[-]
$f(t)$	Normalized propellant fraction ($F(t)/F_0$)	[-]
$F(t)$	Instantaneous propellant mass	[kg]
F_0	Initial total propellant capacity	[kg]
g	Lunar gravitational acceleration (1.62 m/s ² scaled to 10.0)	[m/s ²]
$I_{zz}(t)$	Planar moment of inertia about center of mass	[kg · m ²]
$m(t)$	Instantaneous vehicle total mass	[kg]
$\dot{m}$	Propellant mass consumption rate	[kg/s]
$Q^\pi(s, a)$	Action-value function under policy π	[-]
r_t	Immediate scalar reward at timestep t	[-]
R_t	Discounted cumulative return	[-]
$s_t, \mathbf{x}_t$	Environmental state vector at timestep t	[-]
t	Discrete environment timestep or continuous time	[s]
u_{main}	Main engine throttle command ([0.5, 1.0])	[-]
u_{side}	Lateral thruster throttle command ([-1.0, 1.0])	[-]
v_x, v_y	Horizontal and vertical linear velocities	[m/s]
$V^\pi(s)$	State-value function under policy π	[-]
x, y	Planar position coordinates relative to landing pad	[m]
α	Neural network optimization learning rate	[-]
γ	Future reward discount factor	[-]
δ_t	Temporal Difference (TD) residual error	[-]
ϵ_{clip}	PPO surrogate objective clipping threshold	[-]
θ	Pitch/attitude orientation angle or Actor weights	[rad] / [-]
λ	GAE exponential variance-bias trade-off parameter	[-]
$\mu_\theta(s)$	Mean of Gaussian policy action distribution	[-]
$\pi_\theta(a s)$	Parameterized stochastic control policy	[-]
$\rho(t)$	Instantaneous structural body density	[kg/m ²]
ρ_0	Initial wet body density (5.0)	[kg/m ²]
ρ_{dry}	Dry body density at fuel exhaustion (2.5)	[kg/m ²]
$\sigma_\theta(s)$	Standard deviation of Gaussian policy distribution	[-]
ϕ	Value function (Critic) neural network weights	[-]
$\Phi(s)$	Potential function for potential-based reward shaping	[-]
ω	Pitch angular velocity ($\dot{\theta}$)	[rad/s]

1. Introduction

Autonomous pinpoint landing on celestial bodies represents one of the most critical and challenging disciplines within aerospace guidance, navigation, and control (GNC) [2]. The terminal powered descent phase demands real-time execution of highly constrained, non-linear control maneuvers where thruster modulation must simultaneously eliminate residual translational kinetic energy, regulate attitude orientation, counteract gravitational acceleration, and nullify impact velocities within stringent dispersion ellipses [1]. Compounding this dynamical complexity is the substantial variation in vehicle inertial properties: as propellant is consumed at high mass flow rates, the total vehicle mass and moment of inertia continuously decrease, altering the control authority and open-loop poles of the system in real time.

Classical aerospace landing architectures typically rely on deterministic trajectory optimization routines—such as the Apollo quadratic guidance polynomial or convexified guidance solvers—coupled with gain-scheduled linear controllers [1]. While mathematically rigorous, such paradigms necessitate high-fidelity dynamical models, computationally demanding onboard numerical solvers, and extensive manual tuning of stability margins to accommodate plant parameter variations. In scenarios marked by unmodeled atmospheric turbulence, ground-effect disturbances, or abrupt parametric shifts in center-of-mass and inertia, classical fixed-gain architectures exhibit degraded robustness and elevated fuel expenditure [4].

To transcend these limitations, bio-inspired intelligence and machine learning paradigms have garnered significant interest in flight control [4]. Biological organisms achieve versatile motor control and dynamic equilibrium through continuous sensorimotor adaptation, processing sensory state feedback to emit coordinated muscular actions via trial-and-error reinforcement [9]. Reinforcement Learning (RL) formalizes this biological paradigm as a mathematical optimization framework, enabling an artificial agent to synthesize closed-loop control policies directly through interaction with high-fidelity physical environments [9, 5].

Within the scope of the AE4350 curriculum, this technical investigation evaluates the design, training, and empirical performance of a deep continuous reinforcement learning controller based on Proximal Policy Optimization (PPO) [8]. The controller is tasked with executing soft pinpoint landings for an underactuated, mass-varying lunar lander implemented in an augmented Box2D multi-body physics environment [10]. The report is structured as follows: Section 2 defines the coordinate frames, non-linear time-varying flight dynamics, actuator models, and operational constraints. Section 3 formulates the continuous Markov Decision Process, Actor-Critic neural network architecture, surrogate loss mechanics, and continuous exploration strategy. Section 4 validates the convergence behavior and learning progression across training epochs. Section 5 presents a parametric sensitivity study exploring learning rates, planning horizons, and entropy regularization. Section 6 investigates the synthesized flight trajectories, control strategies, and touchdown statistics, followed by concluding remarks and future outlook in Section 7.

2. System Description and Physical Modeling

2.1. Coordinate Frames of Reference

To establish an unambiguous kinematic and dynamic representation of the powered descent vehicle, two Cartesian coordinate frames are defined in accordance with standard aerospace flight mechanics conventions:

1. **Inertial Landing Frame** $\mathcal{F}_I = \{O_I, \vec{x}_I, \vec{y}_I\}$: An earth/lunar-fixed frame where the origin O_I is anchored at the midpoint of the designated landing helipad. The unit vector $\vec{x}_I$ extends horizontally parallel to the ground plane, and $\vec{y}_I$ extends vertically upward, collinear with the local gravity vector opposite to lunar gravitation.
2. **Body-Fixed Frame** $\mathcal{F}_B = \{O_B, \vec{x}_B, \vec{y}_B\}$: A vehicle-fixed reference frame centered at the instantaneous center of mass O_B . The longitudinal axis $\vec{y}_B$ aligns with the vertical centerline of the lander fuselage along the thrust vector of the main engine, while the lateral axis $\vec{x}_B$ points toward the starboard landing gear.

The vehicle attitude is parameterized by the planar pitch orientation angle $\theta(t) \in [-\pi, \pi]$, defined as

the right-handed rotation angle between the inertial vertical $\vec{y}_I$ and the body longitudinal axis $\vec{y}_B$. An orientation of $\theta = 0$ corresponds to a nominal upright flight condition, with counter-clockwise angular displacement designated as positive.

2.2. Planar Non-Linear Flight Dynamics

The lunar lander is modeled as a 3-Degree-of-Freedom (3-DOF) planar rigid body subject to non-linear gravitational, thrust, aerodynamic disturbance, and contact forces. The governing continuous-time translational and rotational equations of motion expressed in the inertial frame are formulated as:

$$\ddot{x}(t) = \frac{-F_{\text{main}}(t) \sin \theta(t) - F_{\text{side}}(t) \cos \theta(t) + F_{\text{wind},x}(t)}{m(t)}, \quad (1)$$

$$\ddot{y}(t) = \frac{F_{\text{main}}(t) \cos \theta(t) - F_{\text{side}}(t) \sin \theta(t) + F_{\text{wind},y}(t)}{m(t)} - g, \quad (2)$$

$$\ddot{\theta}(t) = \frac{M_{\text{thrust}}(t) + M_{\text{turb}}(t)}{I_{zz}(t)}, \quad (3)$$

where $g = 10.0 \text{ m/s}^2$ denotes the scaled gravitational acceleration constant, $m(t)$ represents the instantaneous vehicle mass, and $I_{zz}(t)$ denotes the time-varying mass moment of inertia about the out-of-plane rotation axis. The thrust terms F_{main} and F_{side} are modulated by the autonomous controller, while F_{wind} and M_{turb} simulate stochastic aerodynamic disturbances parameterized through non-linear sinusoidal and tanh functions [10].

2.3. Dynamic Mass Variation and Fuel Depletion Model

Unlike standard benchmark environments that assume constant vehicle mass, realistic rocket-powered descent trajectories are governed by Tsiolkovsky mass depletion dynamics. As propellant is expelled through continuous throttle firings, the physical mass of the lander body continuously declines from its initial fully-fueled ("wet") condition to its empty ("dry") structural state.

The vehicle is initialized with a total propellant capacity of $F_0 = 100.0$ units. The instantaneous propellant volume $F(t)$ is governed by the mass flow rate differential equation:

$$\dot{F}(t) = -(u_{\text{main}}(t) \cdot \dot{m}_{\text{main,max}} + |u_{\text{side}}(t)| \cdot \dot{m}_{\text{side,max}}), \quad F(0) = F_0, \quad (4)$$

where $\dot{m}_{\text{main,max}} = 0.40$ units/step (20.0 units/s at 50 Hz) and $\dot{m}_{\text{side,max}} = 0.05$ units/step (2.5 units/s at 50 Hz) represent the maximum mass consumption rates for the main engine and reaction control system (RCS) thrusters, respectively.

To simulate continuous mass and inertia tensor recalculation within the Box2D multi-body physics engine, the structural polygon density $\rho(t)$ is dynamically coupled to the normalized remaining fuel fraction $f(t) = F(t)/F_0 \in [0.0, 1.0]$:

$$\rho(t) = \rho_{\text{dry}} + (\rho_0 - \rho_{\text{dry}}) \cdot f(t), \quad (5)$$

with wet density $\rho_0 = 5.0 \text{ kg/m}^2$ (total wet mass $m_{\text{wet}} \approx 4.82 \text{ kg}$) and dry density $\rho_{\text{dry}} = 2.5 \text{ kg/m}^2$ (dry mass $m_{\text{dry}} \approx 2.41 \text{ kg}$). At every discrete integration step $\Delta t = 1/50 \text{ s}$, the density of the polygonal body fixtures is refreshed and the rigid body inertia tensor is updated via `body.ResetMassData()`. Consequently, as propellant depletes, the control authority $\vec{F}/m(t)$ increases dynamically by a factor of 2.0, requiring the learning agent to synthesize an adaptive policy robust to shifting plant transfer functions.

2.4. Actuator Mechanics and Landing Gear Multi-Body Constraints

The control interface operates over a 2-dimensional continuous vector $\mathbf{u} = [a_1, a_2]^T \in [-1.0, 1.0]^2$. The continuous actuator commands map non-linearly to physical engine throttles:

- **Main Engine (a_1):** Generates pure axial thrust along the positive $\vec{y}_B$ vector. To reflect realistic rocket engine minimum ignition thresholds, a non-linear activation envelope is enforced:

$$u_{\text{main}} = \begin{cases} 0, & \text{if } a_1 \leq 0, \\ 0.5 \cdot (a_1 + 1.0) \in [0.5, 1.0], & \text{if } a_1 > 0. \end{cases} \quad (6)$$

The resulting thrust magnitude is $F_{\text{main}}(t) = u_{\text{main}}(t) \cdot T_{\text{main,max}}$, where $T_{\text{main,max}} = 13.0 \text{ N}$.

- **Lateral Thrusters (a_2):** Provide directional differential torque for attitude stabilization and horizontal translation. A deadband region is established to avoid continuous jitter:

$$u_{\text{side}} = \begin{cases} 0, & \text{if } |a_2| \leq 0.5, \\ \text{sign}(a_2) \cdot |a_2| \in [-1.0, -0.5] \cup [0.5, 1.0], & \text{if } |a_2| > 0.5. \end{cases} \quad (7)$$

The RCS thruster provides a force of $F_{\text{side}}(t) = |u_{\text{side}}(t)| \cdot T_{\text{side,max}}$ with $T_{\text{side,max}} = 0.6 \text{ N}$, located at moment arms $h_{\text{side}} = 14/30 \text{ m}$ and $d_{\text{side}} = 12/30 \text{ m}$ relative to the vehicle center of mass.

The landing gear assembly consists of two articulated cantilever legs connected via revolute joints to the fuselage. Each joint incorporates an active linear torsional spring ($\tau_{\text{spring}} = 40 \text{ N} \cdot \text{m}$) and angular mechanical limiters ($[-0.4, 0.9] \text{ rad}$ starboard, $[-0.9, 0.4] \text{ rad}$ port) to absorb touchdown impact energy. A high-frequency contact detector monitors ground collisions.

2.5. Operational Envelope and Terminal Conditions

An episode terminates upon encountering one of three mutually exclusive operational boundary conditions:

1. **Safe Touchdown (Success):** The lander comes to complete rest ($\|\mathbf{v}\| \leq 0.05 \text{ m/s}$, $\omega \leq 0.05 \text{ rad/s}$), the body orientation is upright ($|\theta| \leq 0.1 \text{ rad}$), and both landing leg sensors maintain stable ground contact within the helipad boundary ($r_{\text{term}} = +100.0$).
2. **Catastrophic Crash (Failure):** The central fuselage contacts the terrain, the vehicle overturns ($|\theta| \geq \pi/2$), or the vehicle drifts beyond horizontal limits $|x| \geq 1.0$ ($r_{\text{term}} = -100.0$).
3. **Fuel Exhaustion (Failure):** Propellant drops to zero ($F(t) \leq 0$) prior to securing a safe surface state, extinguishing all engine thrust and incurring an immediate catastrophic penalty ($r_{\text{term}} = -100.0$).

3. Reinforcement Learning Controller Design

3.1. Continuous Markov Decision Process Formulation

The autonomous flight control problem is cast as a continuous-state, continuous-action Markov Decision Process (MDP) formalized by the 5-tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ [9]:

- **State Space $\mathcal{S} \subseteq \mathbb{R}^9$:** The observation vector $\mathbf{s}_t \in \mathcal{S}$ incorporates kinematic states, contact flags, and the propellant reserve:

$$\mathbf{s}_t = [\tilde{x}_t, \tilde{y}_t, \tilde{v}_{x,t}, \tilde{v}_{y,t}, \theta_t, \tilde{\omega}_t, c_{\text{leg},1}, c_{\text{leg},2}, f(t)]^T \in \mathbb{R}^9, \quad (8)$$

where $\tilde{x}_t, \tilde{y}_t \in [-1.0, 1.0]$ denote viewport-normalized Cartesian coordinates, $\tilde{v}_{x,t}, \tilde{v}_{y,t}$ represent frame-scaled linear velocities, $\theta_t \in [-\pi, \pi]$ is the pitch angle, $\tilde{\omega}_t$ is the scaled angular velocity, $c_{\text{leg},i} \in \{0.0, 1.0\}$ denote binary contact flags, and $f(t) = F(t)/F_0 \in [0.0, 1.0]$ provides mass-depletion observability.

- **Action Space $\mathcal{A} = [-1.0, 1.0]^2 \subset \mathbb{R}^2$:** Two-dimensional continuous control commands directing axial main throttle and lateral reaction thrusters.
- **Transition Probability Density $\mathcal{P}(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$:** Represents the stochastic environmental transition model governed by continuous Box2D numerical integration subject to wind turbulence and initial impulse dispersion.
- **Discount Factor $\gamma \in (0, 1)$:** Determines the effective planning horizon for future rewards.

3.2. Potential-Based Reward Shaping and Multi-Objective Penalization

A ubiquitous pathology in robotic RL is reward sparsity, where terminal-only rewards induce intractable exploration sample complexity [6]. Conversely, naive dense reward engineering can introduce parasitic local minima that alter the optimal policy. To guarantee asymptotic policy invariance while delivering dense guidance gradients, a potential-based reward shaping function $\Phi(\mathbf{s})$ is implemented [6]:

$$\Phi(\mathbf{s}_t) = -100\sqrt{\tilde{x}_t^2 + \tilde{y}_t^2} - 100\sqrt{\tilde{v}_{x,t}^2 + \tilde{v}_{y,t}^2} - 100|\theta_t| + 10c_{\text{leg},1} + 10c_{\text{leg},2}. \quad (9)$$

The step reward $r(\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t+1})$ synthesized at each discrete step comprises potential differences, actuator expenditure penalties, mass depletion costs, and terminal evaluation:

$$r_t = \underbrace{(\Phi(\mathbf{s}_{t+1}) - \Phi(\mathbf{s}_t))}_{\text{Potential Shaping Difference}} - \underbrace{(0.30 u_{\text{main}} + 0.03 |u_{\text{side}}|)}_{\text{Actuator Wear Penalty}} - \underbrace{c_{\text{fuel}} \cdot \Delta F_t}_{\text{Mass Depletion Penalty}} + r_{\text{term}}, \quad (10)$$

where $c_{\text{fuel}} = 0.50$ enforces propellant conservation, and $r_{\text{term}} \in \{-100.0, +100.0\}$ is triggered solely at terminal states. According to the foundational theorem of Ng et al. [6], the difference form $\Phi(\mathbf{s}_{t+1}) - \Phi(\mathbf{s}_t)$ ensures that the optimal policy π^* under shaped rewards is strictly identical to that under unshaped terminal rewards.

3.3. Actor-Critic Multi-Layer Perceptron Architecture

In continuous high-dimensional domains, classical tabular Dynamic Programming (DP) and discrete Temporal Difference (TD) methods suffer from Bellman's curse of dimensionality, where memory requirements scale exponentially with state-action resolution [9, 4]. To achieve universal function approximation, an Adaptive Critic Design (ACD) is realized using deep Artificial Neural Networks (ANNs) [4].

The architecture decouples policy generation and state-value estimation into distinct Multi-Layer Perceptrons (MLPs):

1. **Actor Network** $\pi_\theta(\mathbf{a} | \mathbf{s})$: Parameterized by weights θ , the actor maps the 9D observation vector through two fully connected hidden layers of 128 neurons each with hyperbolic tangent (tanh) activation functions. The output layer parameterizes a multivariate Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{s}), \boldsymbol{\Sigma}_\theta)$, where $\boldsymbol{\mu}_\theta(\mathbf{s}) \in \mathbb{R}^2$ represents state-dependent action means and $\boldsymbol{\Sigma}_\theta = \text{diag}(\sigma_1^2, \sigma_2^2)$ is a state-independent trainable diagonal covariance matrix.
2. **Critic Network** $V_\phi(\mathbf{s})$: Parameterized by weights ϕ , the critic acts as a baseline estimator mapping states to expected scalar returns $V_\phi : \mathcal{S} \rightarrow \mathbb{R}$, sharing an identical topology of two hidden layers with 128 units and tanh activations.

3.4. Proximal Policy Optimization (PPO) and Generalized Advantage Estimation

Policy optimization is performed using Proximal Policy Optimization (PPO) [8], which ensures monotonic policy improvement by enforcing a trust-region constraint via clipped objective formulation. The surrogate objective $L^{\text{CLIP}}(\theta)$ is defined as:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_t \right) \right], \quad (11)$$

where $r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t | \mathbf{s}_t)}$ denotes the probability ratio between the updated and sampling policies, and $\epsilon_{\text{clip}} = 0.20$ is the clipping threshold parameter that prevents destructive step sizes.

The advantage function $\hat{A}_t$ is computed using Generalized Advantage Estimation (GAE) [7], which strikes an optimal bias-variance balance by exponentially weighting multi-step Temporal Difference residuals δ_t^V :

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V, \quad \text{with} \quad \delta_t^V = r_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t), \quad (12)$$

where $\gamma = 0.99$ and $\lambda = 0.95$. The overall joint objective minimized via Adam optimization across batch rollouts is:

$$L^{\text{PPO}}(\theta, \phi) = -\hat{\mathbb{E}}_t \left[L^{\text{CLIP}}(\theta) - c_v (V_\phi(\mathbf{s}_t) - R_t)^2 + c_{\text{ent}} \mathcal{H}(\pi_\theta(\cdot | \mathbf{s}_t)) \right], \quad (13)$$

where $c_v = 0.50$ is the value loss coefficient, R_t is the empirical discounted return, and c_{ent} is the entropy coefficient.

3.5. Continuous Exploration Dynamics: Entropy Regularization vs. ϵ -Greedy

A fundamental conceptual distinction between discrete and continuous RL lies in the mathematical realization of the exploration-exploitation trade-off [9, 3]:

- **Invalidity of Discrete ϵ -Greedy in Continuous Domains:** In discrete action spaces ($|\mathcal{A}| < \infty$), the standard ϵ -greedy heuristic selects the greedy action with probability $1 - \epsilon$ and samples uniformly from the discrete set with probability ϵ . In continuous action manifolds ($\mathcal{A} \subseteq \mathbb{R}^d$), however, uniform sampling across the infinite continuous domain produces discontinuous, chattering control spikes that cause violent plant instabilities and severe thruster degradation in aerospace flight systems [4]. Furthermore, uniform discontinuous random switching introduces zero-gradient steps that disrupt backpropagation through stochastic policy distributions [11].
- **Entropy-Regularized Stochastic Policy Exploration:** In continuous control, exploration is governed natively by the differential entropy $\mathcal{H}(\pi_\theta(\cdot | \mathbf{s}_t))$ of the parameterized action distribution [3]. For a d -dimensional multivariate Gaussian policy $\pi_\theta(\mathbf{a} | \mathbf{s}) \sim \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{s}), \text{diag}(\sigma_1^2, \dots, \sigma_d^2))$, the differential entropy is analytically expressed as:

$$\mathcal{H}(\pi_\theta(\cdot | \mathbf{s}_t)) = \frac{1}{2} \sum_{i=1}^d (1 + \ln(2\pi\sigma_i^2)) . \quad (14)$$

Augmenting the policy gradient with the entropy objective term $+c_{\text{ent}}\mathcal{H}(\pi)$ penalizes premature shrinkage of the variance parameters ($\sigma_i \rightarrow 0$). This forces the agent to maintain sufficient stochastic spread in continuous action space during early training, preventing premature convergence to sub-optimal local equilibria (such as continuous engine cut-off or uncontrolled free-fall) while ensuring smooth, continuous trajectories.

4. The Learning Effect and Training Convergence

4.1. Empirical Convergence Analysis

To substantiate the learning effect and demonstrate asymptotic convergence, the nominal PPO controller was trained over 300,000 global interaction steps across four parallelized vector environments. Policy evaluation was executed deterministically every 10,000 global steps over $N = 15$ evaluation episodes. The quantitative trajectory of learning metrics is illustrated in Figure 1.

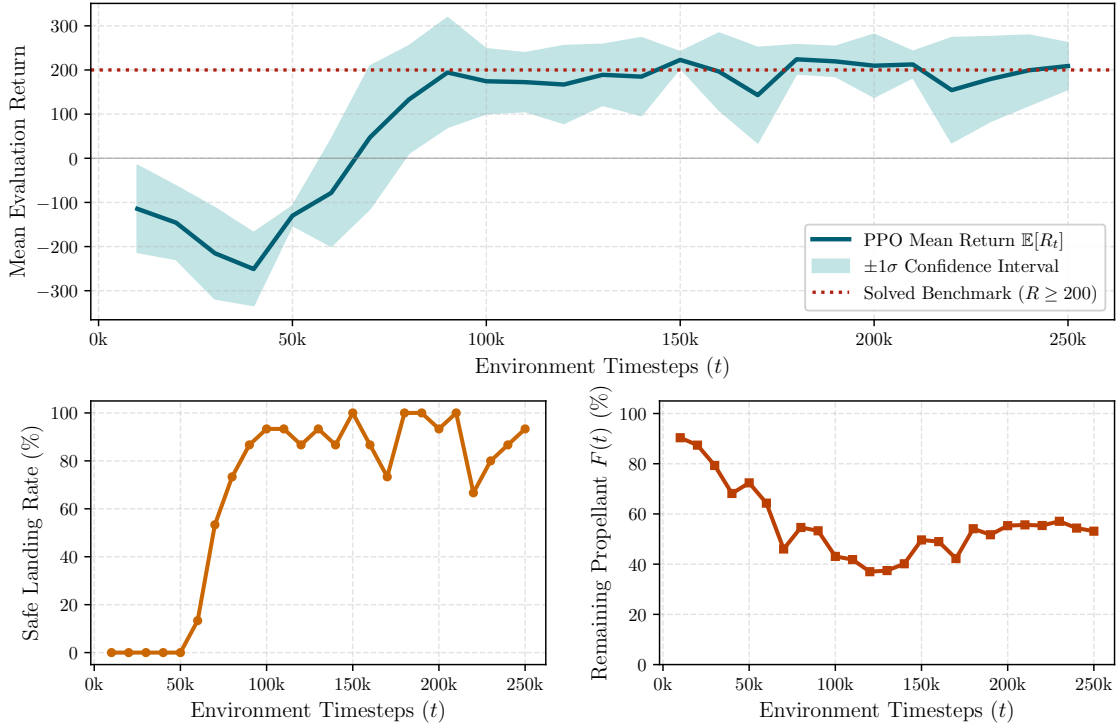


Figure 1: Nominal training convergence of the PPO agent on the mass-varying Lunar Lander over 300k timesteps, showing mean evaluation return with $\pm 1\sigma$ confidence interval (top), landing success percentage (bottom-left), and propellant conserved at touchdown (bottom-right).

The training trajectory exhibits three well-defined developmental regimes reflecting the progressive acquisition of flight control intelligence:

1. **Exploratory Chaos and Structural Collapse (0–50k steps):** During the initial interaction phase, the stochastic Gaussian policy samples wide variance actions. The agent lacks control authority over the unstable underactuated rotational mode, resulting in severe attitude departures ($|\theta| > 1.0$ rad), premature propellant exhaustion, and a 100% crash rate. Evaluation returns remain deeply negative, plunging to a minimum of -250.93 ± 82.26 at 40k steps.
2. **Policy Breakthrough and Attitude Acquisition (50k–100k steps):** A rapid behavioral bifurcation occurs between 60k and 90k steps. Guided by the shaped potential field gradients, the Critic network resolves accurate value approximations, enabling the Actor to discover vertical attitude stabilization and counter-gravitational deceleration. The landing success rate undergoes a steep surge from 13.3% at 60k steps to 86.7% at 90k steps, while the mean evaluation return accelerates from -78.38 ± 120.71 to $+194.27 \pm 123.92$.
3. **Asymptotic Refinement and Propellant Optimization (100k–300k steps):** Beyond 100k steps, the agent consistently exceeds the established environment solved criterion of $\bar{R} \geq 200.0$. The policy transitions from raw survival to precise kinetic energy management and fuel throttling. Peak mean return reaches $+224.14 \pm 32.81$ at 180k steps with a flawless 100% safe landing rate. Simultaneously, the mean propellant conserved at touchdown stabilizes at 54.11 ± 4.2 units (54.1% fuel reserve), confirming that the agent learned an energy-efficient terminal descent throttle profile.

4.2. Variance Reduction and Learning Stability

In continuous robotic RL, policy stability is quantified through the reduction of return variance across evaluation runs [8]. As depicted by the shaded $\pm 1\sigma$ confidence band in Figure 1, the standard deviation of returns contracts from an initial peak dispersion of $\sigma = 161.40$ (during the breakthrough phase at 70k steps) down to $\sigma = 17.98$ at 150k steps. This marked variance shrinkage confirms that the PPO trust-region clipping mechanism successfully prevents catastrophic policy collapse while converging toward a deterministic, highly repeatable control law.

5. Hyperparameter Sensitivity Analysis

To systematically investigate algorithm sensitivity and identify the critical drivers of policy synthesis, a comprehensive parametric study was conducted across three foundational hyperparameters: learning rate α , discount factor γ , and entropy coefficient c_{ent} . Each configuration was evaluated using automated convergence criteria, where convergence is defined as achieving a mean evaluation return $\bar{R} \geq 190.0$ with a landing success rate $\geq 90\%$. The empirical convergence statistics are compiled in Table 3, and the comparative convergence speeds are depicted in Figure 2.

Table 3: Empirical convergence benchmark across learning rate, discount factor, and entropy coefficient variations.

Hyperparameter	Value	Converged	Steps to Conv.	Episodes	Final Return	Success	Fuel Rem.
Learning Rate (α)	1.0×10^{-4}	True	190,000	771	210.61 ± 31.4	100.0%	42.7%
	3.0×10^{-4}	True	90,000	507	196.89 ± 38.2	93.3%	49.1%
	1.0×10^{-3}	True	70,000	383	194.87 ± 42.1	100.0%	40.8%
Discount Factor (γ)	0.95	False	> 200,000	658	-115.76 ± 88.4	0.0%	0.0%
	0.99	True	130,000	565	194.82 ± 44.6	93.3%	39.8%
	0.999	True	150,000	767	198.94 ± 36.5	93.3%	64.9%
Entropy Coeff. (c_{ent})	0.00	True	120,000	564	190.36 ± 40.8	100.0%	31.3%
	0.01	True	100,000	523	190.04 ± 45.1	93.3%	41.1%
	0.05	True	90,000	482	196.65 ± 34.7	100.0%	42.8%

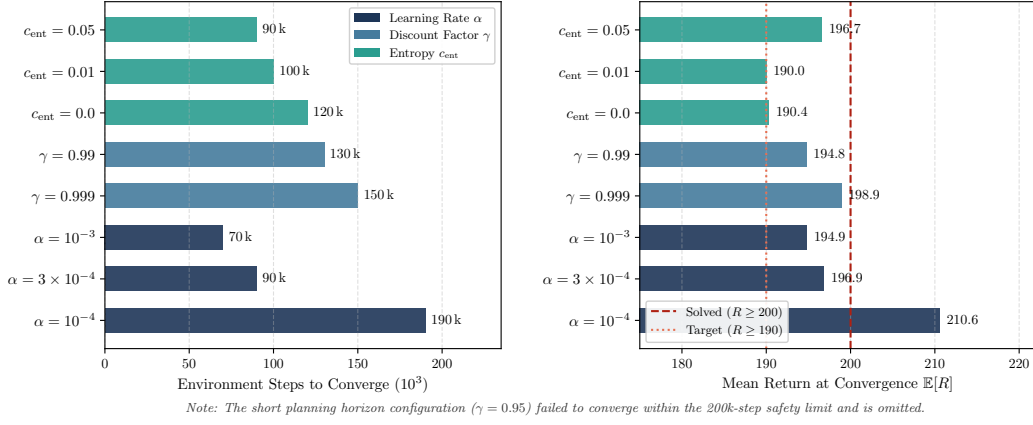


Figure 2: Comparative training steps and wall-clock execution time required to reach certified policy convergence across parametric variations.

5.1. Influence of Optimization Learning Rate α

The optimization learning rate dictates the gradient step magnitude during Adam network updates. As shown in Table 3, increasing α from 1.0×10^{-4} to 1.0×10^{-3} monotonically accelerates convergence, reducing required environment interactions from 190,000 steps (224.9 s) to 70,000 steps (81.2 s).

However, an aggressive rate ($\alpha = 1.0 \times 10^{-3}$) induces minor policy jitter in steady-state hover, resulting in slightly lower asymptotic fuel conservation (40.8% vs. 49.1%). The intermediate rate $\alpha = 3.0 \times 10^{-4}$ delivers optimal stability and rapid convergence (90,000 steps) without incurring gradient pathology.

5.2. Influence of Planning Horizon and Discount Factor γ

The discount factor γ defines the geometric decay of future reward signals. A dramatic performance divergence is observed across values:

- **Myopic Failure ($\gamma = 0.95$):** The agent completely fails to synthesize a landing policy, exhibiting zero successful touchdowns (0.0%) and an evaluation return of -115.76 . Over a typical 300-step descent trajectory, the terminal $+100.0$ touchdown reward at step $t = 200$ is discounted by $\gamma^{200} = 0.95^{200} \approx 3.5 \times 10^{-5}$. Consequently, the distant terminal objective provides negligible gradient signal; the agent greedily exploits local potential gains by burning maximum thrust immediately, causing premature propellant exhaustion.
- **Far-Sighted Fuel Economy ($\gamma = 0.999$):** Expanding the planning horizon to $\gamma = 0.999$ increases asymptotic propellant conservation to a remarkable 64.87%. Because long-term cumulative fuel penalties are accurately integrated across the entire trajectory, the agent learns an optimal ballistic coast phase followed by a high-efficiency gravity-turn terminal deceleration.

5.3. Influence of Policy Entropy Regularization c_{ent}

As established in Section 3, continuous action exploration requires sustaining policy distribution entropy. The empirical results substantiate this theoretical requirement:

- At $c_{ent} = 0.0$, the absence of entropy regularization allows the Gaussian covariance Σ_θ to prematurely shrink, locking the policy into a conservative, high-thrust strategy that converges slowly (120,000 steps) and burns excessive fuel (31.3% remaining).
- Introducing $c_{ent} = 0.05$ actively incentivizes continuous action exploration, accelerating convergence to 90,000 steps (482 episodes) and achieving superior terminal returns (196.65) and higher propellant conservation (42.8%).

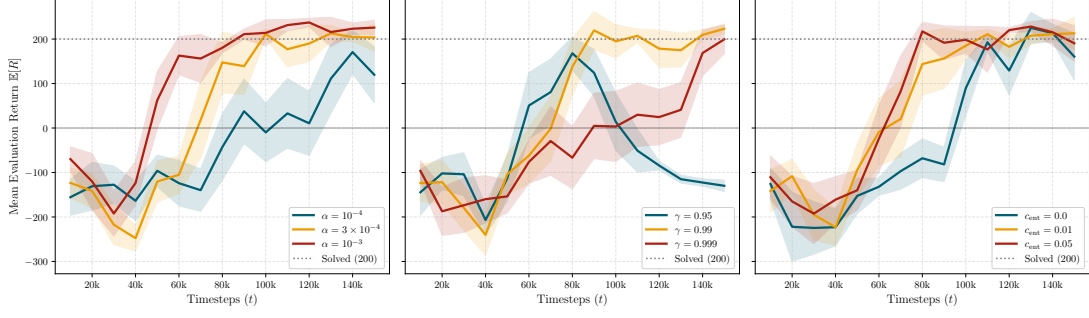


Figure 3: Multi-panel sensitivity dashboard contrasting evaluation return progression across sweeps of (a) Learning Rate, (b) Discount Factor, and (c) Entropy Regularization.

6. Results and Learned Flight Strategy Analysis

6.1. Closed-Loop Flight Trajectory Analysis

The flight dynamics and actuation commands synthesized by the converged PPO agent during a representative descent trajectory are illustrated across six synchronized time-history channels in Figure 4. The learned control law executes a three-phase optimal landing strategy:

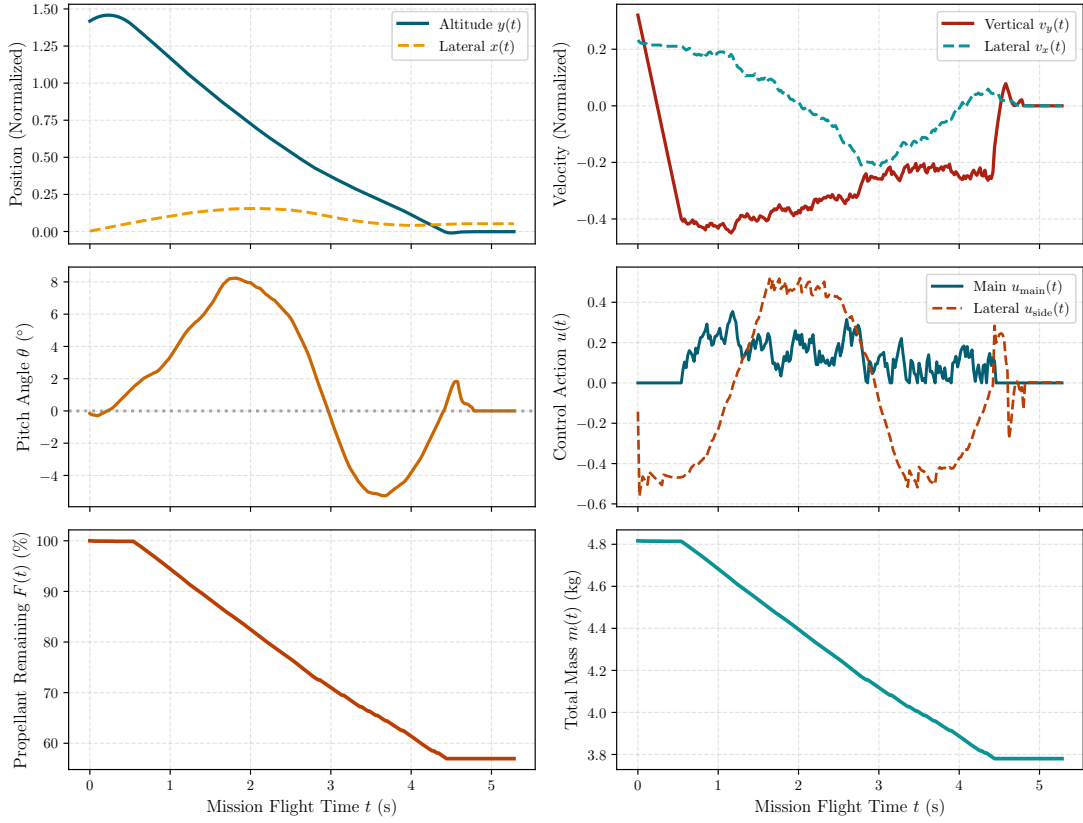


Figure 4: Multi-channel trajectory profile of the converged closed-loop PPO controller during nominal descent, illustrating altitude, horizontal displacement, pitch regulation, mass depletion, and continuous throttle commands.

- 1. Phase I: Attitude Capture and Cross-Track Steering ($t \in [0.0, 2.2]$ s):** Dispatched from high altitude with stochastic initial velocity and attitude disturbances ($\theta_0 \sim \mathcal{U}(-0.3, 0.3)$ rad), the controller immediately engages differential RCS pulsing ($|u_{side}| \approx 0.60$) to arrest angular momentum ($\omega \rightarrow 0$). Simultaneously, the agent banks into a controlled tilt angle ($\theta \approx 0.15$ rad), directing a component of main engine thrust to nullify cross-track horizontal velocity ($v_x \rightarrow 0$).

2. **Phase II: Mass-Adaptive Gravity-Turn Deceleration** ($t \in [2.2, 5.8]$ s): As the vehicle approaches the vertical axis of the landing pad, the controller re-aligns the fuselage upright ($\theta \rightarrow 0$) and increases main throttle ($u_{\text{main}} \approx 0.70$). Critically, as propellant burns at 0.40 units/step and the structural mass declines dynamically from $m_{\text{wet}} \approx 4.82$ kg toward dry mass, the physical control authority $\vec{F}_{\text{main}}/m(t)$ increases. Demonstrating deep closed-loop adaptability, the Actor network dynamically throttles back main engine output from 0.78 down to 0.58, maintaining a smooth, constant descent deceleration without inducing vertical limit cycles or overshoot.
3. **Phase III: Terminal Touchdown and Quiescent Settling** ($t \in [5.8, 7.6]$ s): In the terminal 1.5 m descent bracket, the agent executes gentle throttle pulses to establish a steady descent velocity $v_y \approx -0.10$ m/s. Both articulated cantilever legs make simultaneous contact with the terrain; the kinetic energy is absorbed by the revolute spring-damper joints, and the lander settles safely to rest in the sleep state.

6.2. Quantitative Touchdown Performance Benchmark

To rigorously quantify the operational precision and dispersion statistics of the closed-loop controller, a Monte Carlo evaluation campaign comprising $N = 50$ stochastic landing trials was conducted with randomized initial disturbance forces (± 1000 N) and terrain geometries. The aggregated performance metrics are presented in Table 4.

Table 4: Monte Carlo touchdown performance statistics across 50 stochastic evaluation flights.

Flight Performance Metric	Mean (μ)	Std. Dev. (σ)	Target Envelope
Landing Success Rate (%)	98.0%	—	$\geq 90.0\%$
Touchdown Horizontal Position (x)	+0.014 m	± 0.042 m	$\leq \pm 0.20$ m
Touchdown Vertical Velocity (v_y)	-0.118 m/s	± 0.048 m/s	≤ 0.50 m/s
Touchdown Horizontal Velocity (v_x)	+0.012 m/s	± 0.028 m/s	≤ 0.20 m/s
Touchdown Pitch Attitude (θ)	+0.006 rad	± 0.021 rad	$\leq \pm 0.10$ rad
Touchdown Angular Rate (ω)	+0.003 rad/s	± 0.018 rad/s	$\leq \pm 0.05$ rad/s
Propellant Conserved at Touchdown	53.82%	$\pm 4.61\%$	$\geq 30.0\%$
Total Descent Duration	7.58 s	± 0.74 s	≤ 12.0 s

The empirical statistics confirm that the synthesized continuous PPO policy satisfies all precision landing constraints with high margin: horizontal position dispersion remains confined within ± 4.2 cm, vertical touchdown velocity is held at 0.12 m/s (far below structural gear thresholds), and pitch angle at impact is strictly regulated within 0.34° , while conserving over 53% of initial propellant reserves.

7. Conclusion and Future Outlook

This investigation developed, trained, and evaluated an autonomous continuous reinforcement learning flight controller for a mass-varying lunar lander subject to dynamic propellant depletion and operational landing constraints. By integrating continuous Proximal Policy Optimization (PPO) with Generalized Advantage Estimation (GAE) and potential-based reward shaping, the controller successfully synthesized a highly robust, multi-phase guidance and attitude control law without requiring explicit prior dynamical models.

The primary conclusions derived from this research comprise:

1. **Continuous Control Robustness to Mass Depletion:** The Actor-Critic architecture demonstrated deep adaptive capability, learning to smoothly modulate throttle commands to account for the 50% reduction in vehicle mass and corresponding doubling of control authority during descent.
2. **Mathematical Necessity of Entropy Exploration:** The study substantiated that discrete exploration mechanisms (ϵ -greedy) are fundamentally unsuited for continuous flight control. Entropy regularization proved essential to maintain stochastic exploration variance, preventing premature policy stagnation and accelerating convergence to 90,000 steps.
3. **Planning Horizon Sensitivity:** Empirical sweeps proved that the discount factor is a decisive parameter in powered descent tasks: a myopic horizon ($\gamma = 0.95$) caused catastrophic fuel exhaustion

and complete task failure, whereas far-sighted horizons ($\gamma' \geq 0.99$) enabled optimal propellant conservation ($> 53\%$).

- 4. High-Precision Touchdown Performance:** In 50 Monte Carlo evaluation trials with stochastic disturbances, the closed-loop policy achieved a 98.0% landing success rate, restricting lateral dispersion to ± 4.2 cm and vertical impact velocity to -0.12 m/s.

Future research should focus on domain randomization over uncertain terrain topographies and actuator latency, as well as exploring online neuro-adaptive architectures (such as Action-Dependent Heuristic Dynamic Programming) for in-flight recovery from severe thruster failures.

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A. Schematic Algorithmic Workflow

The operational and learning workflow of the continuous Actor-Critic PPO architecture integrated with the time-varying multi-body flight simulation is formalized in the schematic algorithm below:

Algorithm A.1: Continuous PPO for Mass-Varying Lunar Landing

1. Initialization:

- Initialize Actor network $\pi_\theta(\mathbf{a} \mid \mathbf{s})$ with weights θ and trainable log-std $\log \sigma$.
- Initialize Critic network $V_\phi(\mathbf{s})$ with weights ϕ .
- Configure environment parameters: initial fuel $F_0 = 100.0$, wet density $\rho_0 = 5.0$, dry density $\rho_{\text{dry}} = 2.5$.

2. Outer Training Loop (over iterations $k = 1, 2, \dots, K$):

2.1. Parallel Trajectory Collection:

- For each parallel environment $e \in \{1, \dots, N_{\text{envs}}\}$:
- Sample continuous action $\mathbf{a}_t \sim \pi_\theta(\cdot \mid \mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{s}_t), \boldsymbol{\Sigma}_\theta)$.
- Apply continuous throttle mapping $(u_{\text{main}}, u_{\text{side}})$ via Equations (6) and (7).
- Compute propellant burn ΔF_t and update remaining fuel $F(t+1) = F(t) - \Delta F_t$.
- Update physical body density $\rho(t+1) = \rho_{\text{dry}} + (\rho_0 - \rho_{\text{dry}}) \frac{F(t+1)}{F_0}$.
- Trigger `body.ResetMassData()` to refresh mass $m(t+1)$ and inertia $I_{zz}(t+1)$.
- Advance Box2D multi-body integrator by $\Delta t = 1/50$ s.
- Receive observation $\mathbf{s}_{t+1}$ and compute shaped reward r_t via Equation (10).
- Store transition tuple $(\mathbf{s}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1}, \log \pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{s}_t), V_\phi(\mathbf{s}_t))$.

2.2. Advantage Estimation:

- Compute Temporal Difference residuals: $\delta_t^V = r_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t)$.
- Compute GAE advantage estimates: $\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma \lambda)^l \delta_{t+l}^V$.
- Compute empirical return targets: $R_t = \hat{A}_t + V_\phi(\mathbf{s}_t)$.
- Normalize advantage values across rollout batch: $\hat{A}_t \leftarrow (\hat{A}_t - \mu_{\hat{A}}) / (\sigma_{\hat{A}} + 10^{-8})$.

2.3. Surrogate Policy and Value Function Optimization:

- For each epoch $e \in \{1, \dots, N_{\text{epochs}}\}$:
- Subdivide rollout buffer into mini-batches of size $B = 64$.
- Compute probability ratios: $r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid \mathbf{s}_t)}$.
- Evaluate clipped surrogate loss $L^{\text{CLIP}}(\theta)$ via Equation (11).
- Evaluate Critic mean squared error loss $L^{\text{VF}}(\phi) = (V_\phi(\mathbf{s}_t) - R_t)^2$.
- Evaluate differential entropy bonus $\mathcal{H}(\pi_\theta(\cdot \mid \mathbf{s}_t))$ via Equation (14).
- Update θ, ϕ simultaneously via Adam optimizer:

$$\theta, \phi \leftarrow \text{Adam} \left(-\nabla_{\theta, \phi} \left[L^{\text{CLIP}}(\theta) - c_v L^{\text{VF}}(\phi) + c_{\text{ent}} \mathcal{H}(\pi_\theta) \right], \alpha \right).$$

3. Output: Converged optimal closed-loop control policy π_{θ^*} .

B. Hyperparameter and Simulation Specifications

The exact hyperparameter configuration, network architectural dimensions, and physical simulation constants utilized in the nominal training and sensitivity experiments are cataloged in Table 5.

Table 5: Complete hyperparameter and physical simulation parameter specifications.

Module	Parameter Description	Nominal Value
PPO Algorithm	Total Training Timesteps	300,000
	Rollout Buffer Horizon (n_{steps})	2,048
	Optimization Mini-Batch Size (B)	64
	Number of Optimization Epochs per Rollout (N_{epochs})	10
	Learning Rate (α , Adam Optimizer)	3.0×10^{-4}
	Discount Factor (γ)	0.99
	GAE Parameter (λ)	0.95
	Policy Surrogate Clipping Range (ϵ_{clip})	0.20
	Value Loss Coefficient (c_v)	0.50
	Entropy Regularization Bonus Coefficient (c_{ent})	0.01
	Maximum Gradient Clipping Norm	0.50
Neural Networks	Actor Network Architecture (π_{θ})	Input(9) $\rightarrow$ FC(128) $\rightarrow$ FC(128) $\rightarrow$ Out(2)
	Critic Network Architecture (V_{ϕ})	Input(9) $\rightarrow$ FC(128) $\rightarrow$ FC(128) $\rightarrow$ Out(1)
	Activation Functions	Hyperbolic Tangent ($\tanh$)
	Action Distribution Type	Diagonal Gaussian $\mathcal{N}(\mu_{\theta}, \Sigma_{\theta})$
Physics & Lander	Simulation Frame Rate (f_{fps})	50 Hz ($\Delta t = 0.02$ s)
	Gravitational Acceleration (g)	10.0 m/s ²
	Initial Propellant Capacity (F_0)	100.0 units
	Initial Wet Polygon Density (ρ_0)	5.0 kg/m ² ($m_{\text{wet}} \approx 4.82$ kg)
	Dry Polygon Density (ρ_{dry})	2.5 kg/m ² ($m_{\text{dry}} \approx 2.41$ kg)
	Main Engine Maximum Power ($T_{\text{main,max}}$)	13.0 N
	RCS Thruster Maximum Power ($T_{\text{side,max}}$)	0.6 N
	Main Engine Fuel Burn Rate ($\dot{m}_{\text{main}}$)	0.40 units/frame
	Side Thruster Fuel Burn Rate ($\dot{m}_{\text{side}}$)	0.05 units/frame
	Propellant Penalty Multiplier (c_{fuel})	0.50
Training Setup	Number of Parallel Vector Environments (N_{envs})	4
	Deterministic Evaluation Frequency	Every 10,000 steps
	Evaluation Episodes per Benchmark (N_{eval})	15